Definitions:

Definition 1.1: Let A and B be two sets. We say that A is a subset of B, or that A is contained in B if and only if each x $\in$A is also an element of B. We write A $\subseteq$ B to say that "A is a subset of B." If there is an element of A not in B, we write A $\nsubseteq$ B.

Definition 1.2: The empty set, written $\emptyset$, is the set that has no elements.

Definition 1.3: If A and B are statements, their conjunction is the statement "A and B". A $\wedge$ B is true exactly when both A is true and B is true

Definition 1.4: The intersection S$\cap$T is the set consisting of the elements S and T have in common. One can write S$\cap$T=$x : (x \in S) \wedge (x \in T)$

Definition 1.5: The *disjunction* of two statements A and B is the statement "A or B". A $\wedge$ B is true exactly when either A is true or B is true.

Definition 1.6: The union S $\cap$ T is the set consisting of the elements of either S or T. One can write S$\cap$T= $x : (x \in S) \vee (x \in T$

Definition 1.7: Let T$\subseteq$S be a subset.The complement of T in S is the subset S $\setminus$ T =$x \in S : x \notin T \subseteq S$

Definition 1.8: The converse to the conditional A $\Rightarrow$B is the statement B$\Rightarrow$A.

Definition 1.9: The contrapositive of a conditional A $\Rightarrow$ B is the statement $\overline{\text{B}} \Rightarrow \overline{\text{A}}$

Definition 1.10: We say that two statements A and B are equivalent to mean that A is true if and only if B is true.

Definition 1.11:

Definition2.1: A field is a set F along with operations + and $\cdot$ such that the following hold for a,b,c $\in$ F: Addition: A(0): a+b $\in$ F, A(1):Commutativity, a+b=b+a, A(2): Associativity, a+(b+c)=(a+b)+c, A(3): There is an element 0 $\in$ F such that a+0=a for all a $\in$ F, A(4): There is an element -a $\in$ F such that a + (-a)=0. Multiplication: M(0): a $\cdot$ b $\in$ F. M(1) Commutativity a$\cdot$b = b$\cdot$a M(2) Associativity a$\cdot$(b$\cdot$c)=(a$\cdot$b)$\cdot$c M(4) if a$\neq$0 there exists number 1/a where a$\cdot$1/a =1. M(5) Distributivity,a$\cdot$(b+c)=(a$\cdot$b)+(a$\cdot$c)

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